

A Work Project, presented as part of the requirements for the Award of a Master's degree in
Economics from the Nova School of Business and Economics.

The influence of interest & inflation rate on M&A processes

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Abstract

This thesis analyses the impact of macroeconomic variables such as interest rates and inflation on mergers & acquisitions processes, especially merger waves. ARIMA models are used to first assess the existence and persistence of these waves in the American and European markets, considering values and volumes of deals in previous periods. Two panel data models are included to assess the impact of the macroeconomic variables mentioned above, in which the impact of these variables on both the value and volume of these processes is evaluated. For these analyses data on transactions covering the period from 1977 to 2024 which took place in the United States and in the top five European economies was used.

The results show how the value and volume of transactions from previous periods have an impact on the current period, how high interest rates reduce the value of transactions and how the inflation rate has a negative impact on the volume of transactions.

Keywords: M&A, Interest rate, Merger waves, Inflation

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1. Introduction

Nowadays, most companies are facing a time of uncertainty, mainly caused by inflation and interest rates. After the Covid period and geopolitical emergencies such as the war in Ukraine, companies are left with the question of how to expand and are wondering whether now is the best time to do so. This is due to the fact that despite the 50 basis points cut by the US Federal Reserve, the vast majority of experts expect the current situation to remain unchanged. (World Economic Forum 2024).

In times of uncertainty, i.e. times when there may be high volatility in the markets, companies tend to do less M&A as they usually become more cautious in the sense of investing in growth in an external way (Sgalippa 2017). Companies are also hesitant to execute these processes due to the impact that future regulations may have mainly on the acquired/absorbed company (Emiru and Weisblatt 2024). It can also be argued that these uncertainties can lead to deals taking longer to complete or even to their death (Dang et al. 2022; Gregoriou et al. 2021).

M&A processes are often characterised by clustering, which demonstrates a cyclical pattern, in other words, they can be explained as being heavily influenced by past transact (Ching 2019; Emiru and Weisblatt 2024).

In the economic argot this type of cyclical patterns is known as ‘Merger waves’, many authors have researched about them but there is not a consensual definition and in turn (Harford 2024) it is not clear what causes these phenomena, as many authors usually divide the causes into four categories which will be assessed later (Cho and Chung 2022).

To try to understand and explain these fluctuations, one can use macroeconomic variables such as interest rates, stock prices, inflation rates and others. In this research the macroeconomic variables which will be used will be the interest rate and inflation.

There are different interest rates, but in this case the research will focus on the Key interest rates (Europe) and the closest counterpart which is the federal funds rate (USA). In the United States the Federal Reserve (Fed) is responsible for raising and lowering interest rates. After a 50-bps cut, they want to keep them in a range of 4.75%-5%. (Trading Economics 2024) On the other hand, in Europe the European Central Bank (ECB) is the equivalent of the FED in Europe, and in the year 2024 they have already lowered interest rates twice and are currently at 3.5%. (ECB 2024a; Sanchez 2024)

Central banks use interest rates to control inflation in countries (ESMA 2023). Inflation happens when there is a general increase in prices which leads to a loss in the value of a country's currency over time. This means that people lose purchasing power (can buy less with the same amount of money as prices rise) (ECB 2024b). When inflation is high, central banks raise interest rates, in order to reduce spending and borrowing money. Likewise, when inflation is low, central banks can lower interest rates to increase spending and borrowing. Therefore, from a theoretical point of view, companies would find it more difficult to engage in M&A when interest rates are high because borrowing would be more expensive, which would imply higher financing costs. Likewise, when interest rates are low, firms would be more interested in borrowing money because the cost of financing would be lower.

Since there is no clear motive why Merger waves occur and since it is not fully known how they are affected by the macroeconomic variables mentioned above, the research question arises:

“What influence do inflation and interest rate changes have on M&A processes?”

To answer this question, a review of the literature and perspectives to date will first be conducted to familiarise the reader with the topic. Then data will be collected regarding the

volume and value of M&A processes as well as data on interest rates and inflation since 1977. With these data different economic models will be performed to later analyse and compare the results for the United States with the five largest economies in Europe.

2. Literature

In this section a review of the different research papers on M&A, the different theories on Merger waves, as well as inflation, the interest rate will be conducted.

- **2.1 M&A concept and types**

For the past thirty years, M&A processes have been a subject that has been widely studied (Cartwright and Schoenberg 2006), thanks to this, definitions can be found to explain them. An M&A process is generally understood as a strategy used by companies to consolidate or combine business, with the objective of achieving synergies, expanding to other markets or simply growing. They are an important tool for corporate strategic reorientation and transformation (Jemison and Sitkin 1986).

DePamphilis explains that the concept of Merger and the concept of Acquisition (or tender offer) are not precisely differentiated. He states that mergers usually refers to deals negotiated between the interested parties, i.e. friendly agreements in which two companies decide to join forces. On the other hand, an acquisition occurs when a company makes an offer to the shareholders of another company to acquire their shares at an agreed price or when a firm assumes the control of another company (DePamphilis 2018).

The term acquisition also has other related terms which are essential to understand. Primarily acquisitions can be hostile or friendly.

In friendly acquisitions, the board recommends and agrees to be acquired, and the two companies work together to make the process as smooth as possible. Once the board of the

company to be acquired gives the approval, the company receives a formal offer (called a tender offer).

In contrast, hostile takeovers occur against the wishes of the target company and when the company to be acquired receives an unsolicited offer. This type of case is characterised by the acquirer trying to offer to buy the shares directly from the target shareholder or to buy the shares on the stock market in order to gain control of the company.

It is important to note that hostile takeovers are usually more costly due to the price premium (i.e. the excess of the offer price over the share price of the company) and therefore most takeovers are usually negotiated or friendly takeovers (Ghauri and Buckley 2003; Weston et al. 2014; DePamphilis 2018; Lajoux 2019). When a company buys the total assets or equity of a public company, it is called a buyout. This is a concept that is closely related to the interest rate, as one of the best-known types of buyouts is the so-called LBO (leverage buyout). A leverage buyout is characterised by buying those assets/equity by borrowing money and using those same assets/equity as collateral to guarantee the repayment of the loan requested (Lajoux 2019).

- **2.2 M&A phases & motivations.**

A M&A process is usually divided into different phases. The strategy phase determines the strategy to be used in the M&A process and the type of buyer. In the next phase, the valuation phase, the value of the company to be acquired or sold is evaluated. In the third phase, the financing phase, the financial resources will be obtained in order to be able to realize the acquisition. In the structuring phase, the accounting, financial, and legal choices of the transaction will be planned. The following phase is one of the most important for the deal, the due-diligence phase, which is characterized as the phase in which the possible risks of the transaction are verified and in which it is confirmed that all the data of the acquired company

are true. In the negotiation phase, as the name suggests, the other company is convinced and negotiated to achieve favourable terms for the acquiring company. The closing phase is where the agreement is made official. The last phase, the integration phase, is one of the most critical phases of the whole process, where the two parties work together to achieve a successful transaction (Lajoux 2019).

Authors such as Trautwein and Dephamphilis among others have developed and explained the reasons why M&A processes are driven. The following are the most relevant ones.

One of the most important motivations for this type of process is synergy, which can be divided into Operational, financial and financial. Synergy in an M&A exists when the value of two companies combined is greater than the sum of the two companies individually ($A+B > A+B$) (Seth 1990).

Operational synergy includes economies of scale which helps improve operational efficiency by reducing costs and economies of scope which helps companies diversify their offerings by combining operations.

Financial synergy is characterized by a company's ability to reduce its cost of capital through M&A. Basically the combined company would have greater bargaining power to achieve a lower cost of capital.

Within financial synergies there is a large section called diversification. Thanks to diversification, companies can enter new markets which would significantly help them to reduce their risk and at the same time achieve new growth prospects.

On the other hand, strategic realignment is another key reason why companies are looking to realize these processes. This motive is characterized by acquiring capabilities to be able to adapt

to technological changes, or from the point of view of uncertainty, to be able to adapt to possible regulatory and political changes.

In contrast, the so-called “hubris” refers to managers who overestimate their ability to value the target company. This overestimation often translates into an overpayment of the acquisition price and thus a loss of value for shareholders.

Finally, undervaluation or the purchase of undervalued assets is another major motive, as companies try to buy assets which are below their market value. This motive is usually intensified in merger waves where companies tend to become undervalued (Trautwein 1990; DePamphilis 2018).

- ***2.3 Previous studies on M&A & macroeconomic variables.***

Although this is not a little-studied field, studying the existence and persistence of merger waves and the impact of interest rates and inflation rates on them, with a focus on the mentioned regions, can provide a new perspective on firms' strategies for adapting to uncertainty and into regional differences. Most authors come to a similar conclusion that macroeconomic conditions play an important role in the generation and persistence of M&A waves (Harford 2005, 2024). These already mentioned conditions are among others a low interest rate and a stable or moderate inflation rate, since under these conditions firms have greater certainty about the current and future market situation. On the other hand, situations/markets with a high interest rate or inflation rate would lead to a reduction in the formation of merger waves (Gugler et al. 2012; DePamphilis 2018; Ching 2019; Cho and Chung 2022).

Moreover, authors as Adra et al. (2020) do specify in their research that a contractionary monetary policy can predict losses for acquirers and indicate that large uncertainty about monetary policy is strongly associated with declines in acquirer value. In addition, other authors

(Horn et al. 2021) indicate that after a monetary policy shock, M&A activity decreases, which makes it more difficult to become an acquirer.

In general, most authors do not focus specifically on the impact of inflation and interest rates, and how these together with the volume and value of deals have an influence on M&A processes, leading to the formation of merger waves. This research is different from the rest due to the focus used, the comparison of results between the US and Europe, but mainly due to the analysis of volume and value in previous periods, which is complemented with the impact of the aforementioned economic variables.

3. Methodology

- **3.1 ARIMA models**

In order to evaluate merger waves and understand these time dependencies or cyclical patterns, four ARIMA models were used to analyse whether the values and volumes of past M&A processes have a significant influence on subsequent periods. It was decided to use ARIMA models because they are perfect for stationary and non-stationary time series. Also, this type of models is designed to predict time series with trends and autocorrelation. For the deals-value ARIMAS it was decided to use for both Europe and the United States a (1,1,1) ARIMA, the regression formula for these two models (M1 & M3) will be shown below. It is important to mention that I ran the samples individually, i.e. first I ran the European models and then after saving the results and restarting the program and introducing the dataset again, I ran the US models.:

$$\Delta Y_t = c + \phi_1 \Delta Y_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

In this regression one can observe the first difference of the transaction value represented by ΔY_t , the constant represented by c , the autoregressive coefficient and the moving average coefficient (ϕ_1 and θ_1 respectively) and the white error over time (ϵ_t). On the other hand, for the deals-volume (M2 & M4) ARIMAS it was decided after testing several models to use an ARIMA (2,1,2), the regression formula

for the two models is as follows:

$$\Delta Y_t = c + \phi_1 \Delta Y_{t-1} + \phi_2 \Delta Y_{t-2} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \epsilon_t$$

It is not necessary to re-explain the coefficients/variables and constants as they are the same as in the deals-value model, except that this model includes the changes of the two previous periods and the effect of the errors of the two past periods.

• 3.2 Panel data models

Following the ARIMA models, which as discussed above serve to understand the persistence and existence of merger waves and how the volume and value of previous years influence the current period, it was decided to use two panel data models to understand why merger waves occur. In these two models, macroeconomic variables such as the interest rate and inflation are proposed as explanations regarding possible causes of merger waves.

This type of model allows us to capture both the differences between the two regions (United States and Europe, the latter being represented by the top five economies) and the possible variations over the years (1977-2024).

I develop two regression models for panel data, one for the total value of M&A deals by year and region and one for the total volume of M&A deals by year and region.

Model 5 (hereinafter referred to as **M5**):

This model analyses and evaluates how interest rates and inflation affect the total value of all transactions in each year and region. The formula/regression equation would be as follows:

$$Value_{it} = \alpha + \beta_1 InterestRate_{it} + \beta_2 InflationRate_{it} + \mu_i + \lambda_t + \epsilon_{it}$$

In the following, the variables by which this equation is formed will be explained:

The dependent variable “*Value_{it}*” represents the total value of M&A transactions in region “*i*” in year “*t*”, a high value for this variable would indicate that a large number of transactions took place in that

year and region. The Alpha in this model represents the constant of the model. The variables “*InterestRateit*” and “*Inflationrateit*” are independent variables which represent the interest rate and inflation rate in region “*i*” in year “*t*”. “ β_1 ” and “ β_2 ” represent regression coefficients which explain the marginal effects of interest rates and inflation (respectively) on M&A value. To conclude the model, we can observe in the equation the fixed effect of each region that accounts for possible unobserved factors that are constant over time which is represented by “ μ_i ”. There is also the time effect (represented by “ λt ”) which is common in both regions in year “*t*”, which accounts for possible global factors affecting M&A transactions (e.g. changes in international policies or a global recession such as COVID or the 2008 crisis). Finally, “ ϵ_{it} ” is the random error which represents unexplained variability in the value of transactions for each observation.

On the other hand, Model 6 (hereafter referred to as **M6**) analyses and evaluates how interest rates and inflation affect the total number of transactions in each year and region. The formula/equation would be:

$$Volume_{it} = \alpha + \gamma_1 InterestRate_{it} + \gamma_2 InflationRate_{it} + \mu_i + \lambda t + \eta_{it}$$

In this model the dependent variable *Volumeit* represents the total number of M&A transactions in region “*i*” in year “*t*”, similar to the other model a high value for this variable would indicate that in that year or region there were a large number of completed transactions. As for the independent variables, they are the same as in the previous model so they will not be explained again. “ γ_1 ” and “ γ_2 ” similar to the previous model, are coefficients that measure the impact of the independent variables on the volume of transactions. Finally, the fixed effect, the time effect and the error term represent and explain the same as in the previous model.

It is important to understand that if the interest rate coefficients (“ β_1 ” and “ γ_1 ”) and inflation coefficients (“ β_2 ” and “ γ_2 ”) are positive, it would indicate that an increase in interest rates and/or inflation rate would be related to the value or volume of M&A transactions. On the other hand, if they were negative, it would indicate that there is no relationship.

On the other hand, time effects also play a very important role in the two models since, as explained above, they represent global effects and economic conditions that affect different regions simultaneously. In addition, merger waves are the result of the above-mentioned changes, therefore time effects help to capture the merger wave cycle and help to measure the effects of the interest rate and inflation on transactions in an isolated form.

Furthermore, to understand whether it was more appropriate to use a fixed-effects (FE) or a random-effects (RE) model, a Hausman test was performed to assess and understand whether the random effects are correlated with the explanatory variables of the two models. As will be discussed later in the results section, no conclusive information was obtained from the test. Because of these and to ensure the consistency of the estimators, it was decided to go ahead with fixed effects models.

The SDC's Platinum platform, was used to find and compile the data table on M&A processes that took place between 1977 and 2024. At the beginning, before using the filters which will be mentioned now, more than 500,000 M&A processes between 1977 and 2024 could be found. Therefore, it was decided to use the filter for deals that were made in the USA or in the top 5 European economies. Furthermore, it was accepted to take into account the M&A process for the data table only if the form of the deal was a merger, an acquisition of majority interest or an acquisition of assets. Moreover, another requirement for the dataset was that a deal would only be considered to be added if the acquirer was seeking to own more than 50% of the target firm and if the acquirer owned more than 90% of the target firm after completion of the deal. On the other hand, it was decided to take into account only deals in which the name of both the acquirer and the target could be found (and to exclude those that did not have this information in full) and to eliminate deals which were insignificant due to their value and could affect the analysis it was decided to only add deals with a value of 15 million (US dollar) or more.

Finally, it was decided to eliminate all agreements in which the acquirer or target belongs to the financial sector (SIC 6000-6999). Among those companies classified in this SIC are depository institutions and non-depository credit institutions, security brokers, exchanges, insurance carriers and holding &

investment offices, among others.

After using these filters, the dataset is composed of 34,764 deals. In order to be able to identify where the deal occurred, it was decided to use the nation to which the acquirer belonged. Therefore, of these 34,764 deals, about 22% occurred in Europe (7,819 deals) and 78% in the United States (26,945).

Tables I and II (see appendix) show the industries to which both acquiring and target companies belong. As can be seen in these tables from these 13 industries, the High technology industry is the one in which most deals have been made. For the inflation data it was decided to use the data provided by the Macrotrends website which collects data from the World Bank. For the US interest rate data, the Macrotrends website was used again and in the Federal Funds Rate section, the average yield of each year was selected as the interest rate value. With the European interest rate data there were more complications because for the years prior to 1999 there was no data (as the European Union did not yet exist). It was therefore decided to analyse the number of agreements that took place in the sample (1975-2024) in each country of the top 5 economies in Europe and to select the interest rate for the years 1975-1999 of the country in which the most agreements were made. As can be seen in Table III (see appendix), the country with the most deals was the United Kingdom, so the Bank of England was used as the source for the European interest rate. For the remaining years (2000-2024) the interest rate data provided by the European Central Bank was used.

Once all the datasets were completed, it was necessary to prepare the input to Stata, this consisted of an Excel file which had six columns. These columns were the period, the region, the volume of deals in that year, the value of deals for that year, as well as the interest rate and the value of inflation for that period. The dataset for the ARIMA models consists of 5 columns, similar to the dataset, which was used for the panel data model, with the only difference that the region column was omitted because for the ARIMAS it was decided to split the data by region, otherwise the analysis would not make sense, and the results could not be compared. Once the Excel document mentioned above was imported, the region variable had to be configured to be able to work with it (only with panel data models), as its format was

set to 'string'. To change the type of the variable, the encode command was used and a new variable called region_num was created where each category of the variable 'region' will be converted into a number but keeping the original labels (United States & Europe).

4. Results

The following section presents and interpret the results obtained. I will start with the ARIMA models and continue with the Panel data models.

• 4.1 ARIMA model results

These ARIMA models are used to assess whether the value or volume of deals is related to the volume or value of previous deals, something that would help to understand merger waves their implications, and persistency.

The models are split between Europe and the United States, and within that split there is a model for the value of deals and another for the volume of those deals. The results of the two models for Europe are shown below. Other tests, such as the Dickey-Fuller test and tests to evaluate if the residuals of the models are white noise, can be found in the appendix.

As mentioned above, the model for Europe's deals value (**M1**) is an ARIMA (1,1,1), and its output can be seen below.

Table IV: ARIMA Model (**M1**) (1,1,1) for Deals value in Europe

VARIABLES	(1) deals_value	(2) ARMA	(3) Residual Std. Dev. (σ)
Lagged Deals- Value		0.588*** (0.131)	
Lagged Error Term		-1.000 (606.1)	
Baseline Deals - Value	1.509 (3.015)		47.95 (14,530)
Observations	47	47	47

Standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

As can be seen from the results of the **M1** model, the autoregressive component (AR L1) has a coefficient of 0.588 and is highly significant (p=0.009). This means that an increase of 1 million dollars in the value of deals in the previous period translates into an increase of 0.58 million dollars in the current period. On the other hand, the result of the moving average component (MA L1) is -0.99 but is not statistically significant, implying that shocks in previous periods do not have an impact which is consistent with the current periods' values. Also, the residual standard error is around 48, this represents variability which is not explained by the model. This means that although this model is useful for capturing and understanding basic patterns, it could be improved because there is a large amount of variability in the data which cannot be accounted for by the model and therefore the predictions made with the model have a somewhat high margin of error. This variability may be due to factors that are not included in the analysis.

On the other hand, the model for the volume of deals in Europe (**M2**) is an ARIMA (2,1,2).

Table V: ARIMA Model (**M2**) (2,1,2) for Deals volume in Europe

VARIABLES	(1) deals_volume	(2) ARMA	(3) Residual Std. Dev. (σ)
Lagged Deals- Value (1 st order)		1.068*** (0.408)	
Lagged Deals- Value (2 nd order)		-0.621*** (0.235)	
Lagged Error Term (1 st order)		-0.925** (0.469)	
Lagged Error Term (2 nd order)		0.329 (0.353)	
Baseline Deals- Volume	2.266 (8.054)		50.56*** (4.493)
Observations	47	47	47

Standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

For the second European model (**M2**) the first order coefficient (AR L1) is 1.067 and is statistically significant, which means that an increase of 1 unit in the volume of deals in the previous period is associated with an increase of 1.067 units in the current period. The second order coefficient (AR L2) is also highly statistically significant but in this case, it is negative (-0.62). This coefficient suggests that there is a corrective effect in the subsequent periods, implying that after cycles of large deals volumes, adjustments/corrections would be generated in the following periods. On the other hand, the first-order average component (MA L1) has a negative coefficient (-0.925) and is statistically significant, suggesting that shocks in the previous period have a negative effect on deal volume in the current period. Finally, the residual standard error is 50.56.

As for the US models, below is the ARIMA (1,1,1) (**M3**) output for the deals value as mentioned above.

Table VI: ARIMA Model (**M3**) (1,1,1) for Deals value in USA

VARIABLES	(1) deals_value	(2) ARMA	(3) Residual Std. Dev. (σ)
Lagged Deals- Value		-0.982*** (0.109)	
Lagged Error Term		1.164*** (0.298)	
Baseline Deals - Value	4.609 (15.24)		89.89*** (18.65)
Observations	47	47	47

Standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

In model **M3** it can be observed that unlike model **M1**, the autoregressive component of (AR L1) is negative (-0.98), and highly statistically significant, indicating that an increase in the previous period would produce an average decrease of around 0.9 million dollars in the current period. On the other hand, the moving average component (MA L1) shows a positive value (1.16) and is also statistically

significant, which means that a shock in the previous period has a positive impact in the current period. This can be understood as a recovery effect after certain events in the market. It is worth mentioning that in this model the residual standard error is 89.8, which is higher compared to the **M1** model.

As mentioned above, the model for the volume of deals (**M4**) is an ARIMA (2,1,2).

Table VII: ARIMA Model (**M4**) (2,1,2) for Deals volume in USA

VARIABLES	(1) deals_volume	(2) ARMA	(3) Residual Std. Dev. (σ)
Lagged Deals- Value (1 st order)		0.753*** (0.199)	
Lagged Deals- Value (2 nd order)		-0.713*** (0.189)	
Lagged Error Term (1 st order)		-0.577 (454.8)	
Lagged Error Term (2 nd order)		1.000 (1,577)	
Baseline Deals- Volume	2.876 (30.34)		117.7 (92,847)
Observations	47	47	47

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

On the other hand, the **M4** model presents similar results to the **M2** model. In this case the first order autoregressive coefficient (AR L1) is 0.75 (and statistically significant), indicating that an increase in the previous period would cause an average increase of 0.7 units in the current period. Likewise, the second order coefficient (AR L2) is also statistically significant, but negative (-0.71) which represents a correlation effect. This means that the volume of deals in the current period will be reduced if in past periods there was a large volume.

Finally, the moving average components (MA L1 & L2) are not statistically significant, which implies that possible shocks in previous periods do not have a consistent impact on deal volume.

As can be seen, the results of the **M2** and **M4** models are quite similar, as both have a positive trend in the short term (AR L1) and a cyclical fit over time. However big differences can be found when comparing between model **M1** & **M3**. In the European model (**M1**) it can be observed that the AR L1 indicates a positive inertia, which reflects an increase in the value of the agreements in relation to the previous period. This result shows a more stable market, which thanks to strict regulations is more predictable where high values tend to be persistent. On the other hand, the coefficient of the AR L1 of the **M4** model is negative, which implies a reduction in the value of the agreements in relation to the previous year as discussed above. This cyclical pattern shows a higher volatile where rapid adjustments are found after periods of high activity. This might be stimulated by macroeconomic factors such as the interest rate or simply because the market is less regulated than the European market, which may also have a lower risk tolerance.

In the following, after showing that merger waves are persistent, the results of the panel data models which were conducted to observe how interest rate and inflation affect the M&A process and whether they are the cause of merger waves, will be interpreted and analysed.

• **4.2 Panel data model results**

The table below shows all the variables with which the panel data model was constructed.

Table VIII: Panel data model variables

Variable name	Storage type	Display format	Value label	Variable label
period	int	%10.0g		period
region	str13	%13s		region
deals_volume	int	%10.0g		deals_volume
deals_value	double	%14.2fc		deals_value
interest_rate	double	%10.0g		interest_rate
inflation_rate	double	%10.0g		inflation_rate
region_num	long	%13.0g	region_num	

Once the variables had been verified, the panel data model was started by first running the regressions of **M5** (Value of Deals).

First, the regression was performed with a fixed-effects model and a random-effects model, to assess which type of model should be used, the Hausman test was chosen. Since the results of the Hausman test were inconclusive (see appendix), it was decided to continue with fixed effects models only. These fixed effects models were appropriate when there are region-specific unobserved factors which may be related to the interest rate and inflation

The following table is the output for the **M5** model:

Table IX: Panel data model (**M5**) deals value

	(1)	(2)	(3)
	deals_value	deals_value	deals_value
interest_rate	-17.41*** (3.235)		-13.76** (4.390)
inflation_rate		-23.86*** (5.474)	-8.733 (7.119)
Constant	244.7*** (21.65)	243.0*** (24.77)	257.9*** (24.14)
Observations	96	96	96
Adjusted R-squared	0.221	0.152	0.225
Standard errors in parentheses			
* p<0.05, ** p<0.01, *** p<0.001			

As can be seen, the **M5** panel data model (deals value) (see table IX) is controlled for heterogeneities specific to the two observed regions, which allows us to assess mainly the relationships within each region over the years. The rho value (0.4871) shows that about 49% of the total variance of the deals values is mainly due to differences between the observed regions. This result supports the choice of a fixed effects model as it is clear that fundamental differences between regions (e.g. economic, cultural, regulatory etc.) are important in explaining the value of deals.

The results (see table IX) show that the interest rate has a negative and statistically significant effect on the value of deals ($\beta = -13.757$ $p=0.002$). Taking this into account, it is understood that if the interest rate increases by one percent, the average value of deals will decrease by about 14 million dollars.

On the other hand, although the inflation rate is not statistically significant ($p=0.223$), its coefficient

indicates a negative relationship ($\beta=-8732$).

The R-squared (or coefficient of determination) indicates that about 25% of the variation in deal values within regions is explained by inflation and interest rates, and it can be understood that there are other relevant factors (e.g. local regulations, external economic shocks or business confidence) which are not included in the model. Taking into account the results of the model (M5) it can be stated that inflation has a bigger negative impact on the value of deals when it is evaluated on an individual basis (around 24 million reduction for every 1% increase in inflation). This statement demonstrates how economic uncertainty in the market which is associated with inflation affects firms' strategies. It can also be seen how the interest rate has a consistent negative effect on the value of deals both individually and in combination. On the other hand, due to the correlation between the interest rate and inflation variables, (see test in the appendix) this may not be statistically significant in the combined model.

In general terms, it is important to understand that both rates are of great importance for the value of deals, but that this depends very much on the economic context of the market.

The results of this first panel data model show expected results and reflect the importance of the interest rate in the field of mergers and acquisitions, since it is understood that as interest rates increase, borrowing cost becomes more expensive, which implies that companies lose motivation to make deals. In addition, high interest rates not only increase the cost of financing, but also reflect a restrictive monetary policy, which can lead to rejection by investors. On the other hand, as for the inflation rate, these results could be explained by several reasons, one of them being that companies adapt their different strategies to inflation in order to reduce the impact it has on deals.

The regression of **M6** (Volume of deals) was then performed and the same steps were followed as for the model **M5**. The following table is the output of model **M6**.

Table X: Panel data model (**M6**) deals volume

	(1)	(2)	(3)
	deals_volume	deals_volume	deals_volume
interest_rate	-25.70*** (5.990)		-10.70 (7.864)
inflation_rate		-47.62*** (9.414)	-35.85** (12.75)
Constant	493.1*** (40.09)	535.9*** (42.60)	547.5*** (43.25)
Observations	96	96	96
Adjusted R-squared	0.147	0.199	0.206
Standard errors in parentheses			
* p<0.05, ** p<0.01, *** p<0.001			

As can be seen in the output of the hausman test (see appendix) for **M6**, no conclusive results were obtained either. Because of this, the same decision was taken to continue with a fixed effects model only since, as can be seen in the two Hausman tests, a negative chi2 was obtained.

As for the **M6** (deals volume) model (see table X) it can be seen that the rho value is higher than in the previous model (0.5571) which means that about 56% of the variability in deals volume is mainly due to differences between regions. As in the **M5** model, these results support the choice of a fixed effects model. On the other hand, in relation to the coefficients, the inflation rate has a negative and highly statistically significant effect ($\beta = -35.87$ $p=0.006$) on the volume of deals. This indicates that a one percent increase in the inflation rate will decrease the volume of deals by about 36 deals.

On the other hand, as in model **M5**, the coefficient associated with the interest rate is also negative ($\beta = -10.70$ $p=0.177$), but in this case it is not statistically significant. This means that there is no clear evidence to conclude that the interest rate influences the volume of agreements.

Furthermore, the constant of the **M6** model is 547.525, which means that if the interest rate and inflation were 0, the average volume of agreements would be around 548. As in the **M5** model, inflation also has a negative effect when valued individually. The results show that a 1% increase in inflation would reduce the number of agreements by 48 units. On the other hand, the interest rate as discussed above is not significant (but negative) when valued in combination, which changes when the variable is valued

individually in the model (negative and significant impact). Due to the correlation discussed above, it can be observed that although the interest rate directly influences financing costs, its impact seems to be smaller in relation to the volume of deals than inflation.

These results demonstrate the great importance of the interest rate in M&A processes and how the inflation rate can generate uncertainty and could discourage firms from entering into M&A deals.

In general terms, the results of these two models are very interesting, and even more so if you compare the models with each other. As mentioned above, the value of the deals shows a higher sensitivity to changes in the interest rate, while on the other hand the volume of deals is more influenced by the inflation rate. This difference in the results may be due to several reasons, one of them being that because the interest rate is related to the cost of capital, an increase in the interest rate would imply that financing larger deals is more expensive and because of this firms may be avoiding making larger deals. In addition, inflation affects the firm's overall operating costs and expected revenues, which could mean that firms would be more cautious about entering into such deals.

To conclude with the results of these two models, it can be understood that these results suggest that firms respond differently to economic conditions depending on whether they prioritise the quantity or number of deals made or their size.

5. Limitations

Although this research has been able to demonstrate very important results, it is not without limitations. The sample size only takes into account M&A deals from 1977 onwards, while this provides sufficient data for the analysis to draw conclusions, earlier data could provide more information on merger waves and their causes. Another limitation is the representation of Europe. In this analysis it was decided that Europe would be represented by the five most important economies in the region, which makes the data easier to handle and work with but may limit the results for the region. It was also decided to use the United Kingdom interest rate for Europe until 1999 because Europe did not exist. Although this decision is a reasonable approximation because the largest number of European settlements take place in the UK,

this analysis may not be capturing the full variability of the interest rate in the region before the creation of the ECB. Furthermore, another limitation is the clear correlation found between the macroeconomic variables used in this research, although it shows real dynamics, it is very difficult to isolate the individual effects of these variables in relation to M&A processes. It should also be noted that due to the filters used to maintain consistency in the dataset, a large number of deals have not been analysed. This decision may miss important and significant trends, especially in times of economic downturns. Finally, it should be noted that the models used assume that the historical data used for the analysis are indicative of future trends. This is not a negative thing as such but external shocks or new regulations could affect/limit the results.

6. Conclusion

In this thesis it has been analysed on the one hand how the volume and value of past M&A transactions affect current period deals, and on the other hand the relationship of inflation and interest rates with merger waves. As discussed above, it was decided to use the US and European markets (which is represented by the five major economies) for this analysis. The results show important differences between the economic variables and their relationship with the volume and value of transactions.

Thanks to the analysis of the ARIMAS models in this research, the existence of cyclical patterns (and persistence of these cycles), known as merger waves, has been confirmed. Having analysed the two markets separately, it has been observed that the American market tends to be more volatile than the European market, which may be due to differences in regulations, market risk tolerance and the general economic context. Overall, these results support the idea that the specific characteristics of each region influence companies' strategies in relation to responding to economic market conditions.

On the other hand, thanks to the results of the panel data models, it has been possible to confirm that interest rate and the inflation rate are important drivers of merger waves. It can also be observed that the interest rate has a greater influence on the value of deals and inflation on the volume of deals. These

results show that the cost of capital (directly related to the interest rate) decreases higher value deals, and that inflation generates a state of uncertainty in the market, which reduces the number of deals done.

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8. Appendix:

Variables:

Summary statistics					
	N	Mean	SD	Min	Max
deals volume	96	362.125	340.746	0	1420
deals value	96	155.921	185.610	0	748.576
interest rate	96	5.097	4.382	0	16.39
inflation rate	96	3.65	2.689	-.36	13.55

Industries (Table I & II):

Acquiror Industry	Value
Consumer Staples	1.866
Materials	2.791
Retail	1.768
High Technology	6.831
Energy and Power	4.139
Industrials	4.602
Healthcare	4.074
Consumer Products and Services	3.203
Financials	537
Media and Entertainment	3.466
Telecommunications	1.456
Government and Agencies	25
Real Estate	6
Total	34.764

Target Industry	Value
Consumer Staples	1.881
Materials	2.869
Retail	1.697
High Technology	6.581
Energy and Power	4.333
Industrials	4.899
Healthcare	4.115
Consumer Products and Services	2.718
Financials	609
Media and Entertainment	3.452
Telecommunications	1.523
Government and Agencies	84
Real Estate	3
Total	34.764

Deals Europe Analysis (Table III):

Germany	688	9%
France	953	12%
United Kingdom	4878	62%
Italy	682	9%
Spain	618	8%
Total	7819	

Deals Analysis:

United States	26.945	78%
Europe	7.819	22%
Total	34.764	

ARIMA dickey fuller deals value Europe:

. dfuller deals_value

Dickey-Fuller test for unit root Number of obs = **47**
Variable: **deals_value** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-3.286	-3.600	-2.938	-2.604

Mackinnon approximate *p*-value for Z(t) = **0.0155**.

ARIMA dickey fuller deals volume Europe:

. dfuller deals_volume

Dickey-Fuller test for unit root Number of obs = **47**
Variable: **deals_volume** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-1.830	-3.600	-2.938	-2.604

MacKinnon approximate *p*-value for Z(t) = **0.3656**.

. dfuller d_deals_volume

Dickey-Fuller test for unit root Number of obs = **46**
Variable: **d_deals_volume** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-5.364	-3.607	-2.941	-2.605

MacKinnon approximate *p*-value for Z(t) = **0.0000**.

ARIMA dickey fuller deals value USA:

. dfuller deals_value

Dickey-Fuller test for unit root Number of obs = **47**
Variable: **deals_value** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-2.129	-3.600	-2.938	-2.604

MacKinnon approximate *p*-value for Z(t) = **0.2330**.

. dfuller d_deals_value

Dickey-Fuller test for unit root Number of obs = **46**
Variable: **d_deals_value** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-7.572	-3.607	-2.941	-2.605

Mackinnon approximate *p*-value for Z(t) = **0.0000**.

ARIMA dickey fuller deals volume USA:

. dfuller deals_volume

Dickey-Fuller test for unit root Number of obs = **47**
Variable: **deals_volume** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-1.592	-3.600	-2.938	-2.604

Mackinnon approximate *p*-value for Z(t) = **0.4875**.

. dfuller d_deals_volume

Dickey-Fuller test for unit root Number of obs = **46**
Variable: **d_deals_volume** Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-4.654	-3.607	-2.941	-2.605

Mackinnon approximate *p*-value for Z(t) = **0.0001**.

ARIMA White noise tests deals value Europe:

Portmanteau test for white noise		
Portmanteau (Q) statistic =		11.8233
Prob > chi2(21) =		0.9443

ARIMA White noise tests deals volume Europe:

Portmanteau test for white noise		
Portmanteau (Q) statistic =		19.3475
Prob > chi2(21) =		0.5629

ARIMA White noise tests deals value USA:

Portmanteau test for white noise		
Portmanteau (Q) statistic =		18.1644
Prob > chi2(21) =		0.6386

ARIMA White noise tests deals volume USA:

Portmanteau test for white noise		
Portmanteau (Q) statistic =		32.3082
Prob > chi2(21) =		0.0545

Hausman test fixed & random effects deals_value:

```
. hausman fe_model re_model
```

	Coefficients			
	(b)	(B)	(b-B)	$\sqrt{\text{diag}(V_b - V_B)}$
	fe_model	re_model	Difference	Std. err.
interest_r~e	-13.757	-17.76458	4.007573	.
inflation_~e	-8.732565	-4.257331	-4.475234	.

b = Consistent under H0 and Ha; obtained from **xtreg**.

B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        = -1.88
```

Hausman test fixed & random effects deals_volume:

	Coefficients			
	(b)	(B)	(b-B)	$\sqrt{\text{diag}(V_b - V_B)}$
	fe_model	re_model	Difference	Std. err.
interest_r~e	-10.7038	-18.96499	8.261184	.
inflation_~e	-35.85038	-26.62516	-9.225218	.

b = Consistent under H0 and Ha; obtained from **xtreg**.

B = Inconsistent under Ha, efficient under H0; obtained from **xtreg**.

Test of H0: Difference in coefficients not systematic

```
chi2(2) = (b-B)'[(V_b-V_B)^(-1)](b-B)
        = -1.85
```

Correlation test interest & inflation rate (panel data models):

```
. pwcorr interest_rate inflation_rate, sig
```

	intere~e	inflat~e
interest_r~e	1.0000	
inflation_~e	0.6747	1.0000
	0.0000	

